System Overview · language-guided multi-floor UAV search

帮我看看二楼厨房台面附近的微波炉、二楼客厅沙发旁的茶几和三楼客厅的电视是不是都还在原位，再顺路确认一下二楼卫生间马桶旁还有没有厕纸，先后顺序你自己安排。

Qwen semantic grounding · no prescribed order

L2

Microwave

L2 · kitchen
near countertop

L2

Coffee table

L2 · living room
near sofa

L3

Television

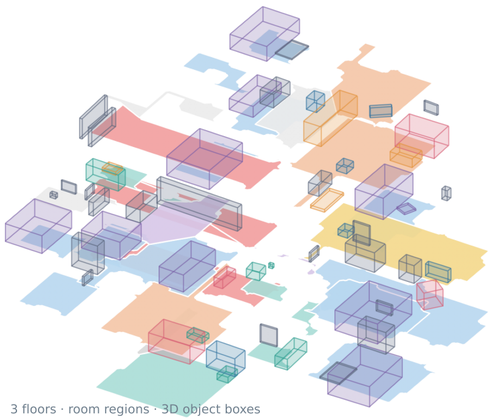
L3 · living room
instance

L2

Toilet paper

L2 · bathroom
near toilet

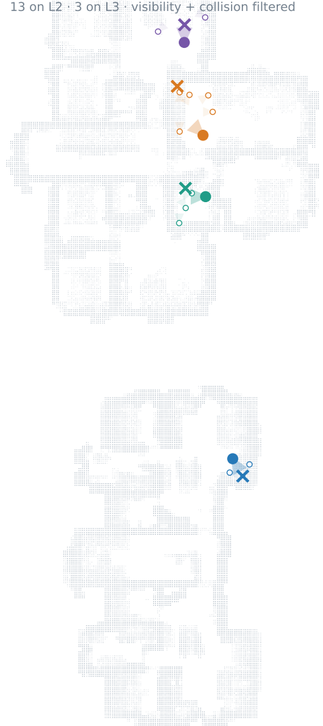
① Pre-explored semantic memory



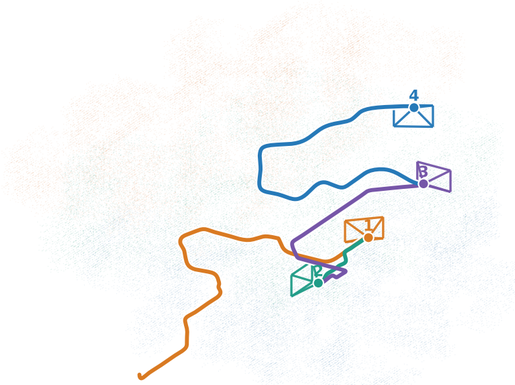
3 floors · room regions · 3D object boxes

② 16 feasible observation poses

13 on L2 · 3 on L3 · visibility + collision filtered



③ Joint task–viewpoint–path optimization



16 feasible views · 48.1 m · 3D A* + B-spline

④ Online execution and visual verification

real RGB · OWLv2 · 4/4 targets confirmed

